

Communication Requires Separation: The Causal Structure of Cell Signalling

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Abstract

Biologists distinguish a class of causal relations within and between cells as signalling, and treat the criteria for doing so as self-evident. The paper analyses the concept to supply those criteria. It is argued that signalling requires detachment: causal separation within and between the processes involved. Detachment grounds three criteria, individually necessary and jointly sufficient. Signals are transduced at the point of reception, contributing to the response without being incorporated into it. Reception is active, the receiver producing its response through mechanisms it maintains independently of the signal. The sender is sensitive to that response, and is sustained by it. The account is developed through two concurrent processes in *Bacillus subtilis* biofilms, glossed by the researchers who study them as metabolism and as communication respectively. In the literature, signalling has been characterized as the absence of material continuity and as minimal causation, and identified by organizational closure; each of these fails to discriminate the pair. Detachment does, while allowing the physical properties of a signal vehicle to carry explanatory weight. The results bear on the debates on biological information, constraining the interpretation of sender-receiver models and the analysis of biological control.

Keywords: biological information; causal specificity; sender-receiver models; bacterial communication; biological autonomy; causal pathways; detachment; biological control

1 INTRODUCTION

Causal processes in living systems are widely classified in informational terms. DNA fragments are said to *code for* proteins and traits, neural activity to *represent* a stimulus. The commitments of these claims are rarely stated explicitly by scientists—and the scope and justification of informational language in biology remain a point of contention for philosophers.

This question takes a specific form in cell biology. A special class of causal relations within and between cells is described as *signalling*. Indeed, cell signalling has become a crucial concept in contemporary biological research, taking on an explanatory role in many circumstances (see Bradshaw and Dennis 2004; Hancock 2017). Yet, the criteria for distinguishing signalling from other causal relations are taken as self-evident and missing from published reports. The intuitive account draws on the formal, higher-level concept of communication (Shannon and Weaver 1964), casting signalling as a process in which a sender transmits a signal over a channel to a

receiver. But these engineering criteria are too weak on their own. Transmission of airborne disease satisfies this structure, and it is not signalling.

The goal of the paper is to provide a causal characterization of cell signalling, and the characterization offered is that communication requires separation: a signal must reach a receiver that is causally detached from it. The characterization should allow for identifying under what criteria biologists apply the concept of signalling instead of a straightforward, causal-mechanical frame, for instance distinguishing signalling from metabolism, or from disease transmission.

An intuitive characterization is that signals are “removed from immediate physical effects.” This intuitive notion of “removal” draws from the folk concept of communication. If I see my friend accidentally stepping off the pavement into approaching traffic, I can either communicate the danger by shouting “Watch out, a car!” or just pull them back onto the pavement. In the first case, when communication occurs, their stepping back on the pavement is not a physical effect of my actions in the same sense as in the second case, when no communication happens. However, in both cases there is definite causal connection between my actions and them stepping back.

To substantiate the notion of “removal”, I characterize signalling in terms of detachment, causal separation within and between processes. Detachment is a graded concept and unevenly distributed through a process, so it does not by itself identify where signalling begins. However, its presence in three components of the causal process is crucial. Thus, it provides a framework to specify three criteria that are individually necessary and jointly sufficient for signalling:

- (i) transduction: signals are materially discontinuous in relation to responses,
- (ii) active reception: the mechanisms which allow the recipient system to produce a response are maintained independently of the signal, and
- (iii) sender-receiver asymmetry: the sender is sensitive to the receiver’s response, and that response contributes to the sender’s maintenance in a way the sender cannot secure by itself.

These criteria have been previously stated independently and in different contexts, primarily in relation to animal communication (see Scott-Phillips 2008; Artiga, Birch, and Martínez 2020, and other papers in that special issue; see also Artiga 2021b; Skyrms 2010), but have not been explicitly considered in relation to cell signalling (important exceptions being Calcott, Pocheville, and P. Griffiths 2020; Godfrey-Smith 2014). The concept of transduction relates to characterization of signalling in terms of causal pathways (see Boniolo and Campaner 2018; Ross 2018, 2021a; Thagard 2003) and the related distinction of material and signal continuity (Ross 2021b). It also figures strongly in biosemiotic accounts of signalling (e.g., Barbieri 2008). The notion of active reception has been proposed by Arnon Levy (2011) when discussing biological information, though he considered the activity as metaphorical. My characterization also develops Marc Artiga’s notion of signals as non-enabling causes (Artiga 2021b, 2021a), which in turn built on J.B.S. Haldane’s (as quoted by Wilson 1975) and R. Haven Wiley’s (1994) view of signals as energetically cheap compared to responses they elicit. The most direct sustained engagement with the question of what distinguishes signalling is in the work of Howard H. Pattee (in particular, Pattee [1969] 2012). Pattee developed a holistic view of simple signals as operating within a switching network, and defined by that network. Pattee’s proposal has been recently explored in the context of the organizational approach to biology (Frick, Bich, and Moreno 2019) which explicitly considered the use of “signalling” in synthetic biology.

In the current paper, I build on these existing threads, interlacing them with the causal analysis of signalling within the interventionist framework (Woodward 2004, 2010), to establish

clear criteria for the signalling concept at the cellular level. In this way the current account goes beyond the existing causal analysis by Calcott and colleagues (2020), which assumed signalling to be present in the analysed cases.

As I will argue based on case studies, neither of the previous accounts is independently sufficient to distinguish cell signalling from non-signalling interactions. Only when their criteria are developed jointly, viewed as properties of the causal structure of a process under consideration, they offer a way of delineating signalling in simple biological systems.

The paper is structured as follows. In [section 2](#), I review two sets of experiments on bacterial biofilms which offer a useful testing ground for concepts of signalling—they consider two concurrent processes through which metabolism is coordinated in the colony, with one being described in purely metabolic, and the other signalling terms. In [section 3](#), I offer an interventionist analysis of the case studies and show that earlier proposals fail in adequately distinguishing the two frames. Based on this analysis, in [section 4](#), I explore the concept of detachment which offers an overarching framework for the criteria to be applied jointly, and discuss the criteria in detail. In [section 5](#), I show how the offered account of signalling fits into the debates on biological information, specifically focusing on sender-receiver models, causal specificity, and control in biological systems, which variously rely on the notion of signalling.

2 SIGNALLING AND METABOLISM IN BACTERIAL BIOFILMS

Two sets of experiments on *Bacillus subtilis* biofilms conducted at Gürol M. Süel's lab at the University of California, San Diego offer a clear example of the nuances in distinguishing signalling from other kinds of cellular processes. *B. subtilis* is a model organism (for discussion of *B. subtilis* as a model, see Barák 2021; Errington and Aart 2020; for model organisms see Ankeny and Leonelli 2020) used, among others, to study biofilms, a form of cross-feeding community of bacteria. The two sets of experiments I wish to focus on (reported in Liu et al. 2015; Prindle et al. 2015) centre on the coordination of growth and the synchronization of metabolism between individual bacteria making up the biofilm. They highlight two concurrent processes:

1. Bacteria at the centre of a growing biofilm start to deplete the nutrients, and the amount of ammonium they produce and diffuse throughout the biofilm starts to decrease. This results in the halting of cell division of bacteria on the periphery, slowing down the growth of the colony until the nutrients available on the outside can diffuse to the centre (see Liu et al. 2015).
2. Bacteria at the centre of a growing biofilm start to deplete the nutrients and in response depolarize by releasing potassium, which is actively propagated throughout the biofilm, until it reaches the periphery. The depolarization wave limits the ability of those cells to take up and retain nutrients, allowing nutrients to diffuse through the biofilm and reach the cells in the centre, effectively synchronizing the metabolic states of cells across the biofilm (see Prindle et al. 2015).

There are clear causal connections between processes characterized as metabolic co-dependence and as signalling. Due to their intertwinement (indeed, they are also characterized as a coupled system by Prindle et al. (2015)) and the differences in adopted frames, these two processes offer a useful testing ground for accounts of cell signalling: any appropriate account of signalling should classify the second, but not the first as an instance of communication.

2.1 *Metabolic co-dependence*

In the first case, the researchers describe the process directly in causal-mechanical terms (following the account in Liu et al. 2015): ammonium is a substrate required for the production of glutamine (essential for growth) in a process where an enzyme, glutamine synthetase, combines it with glutamate (Gunka and Commichau 2012). Ammonium can be produced by the bacterium itself from glutamate in the form of ammonia vapor. However, the vapor is in equilibrium with ammonium and can cross the cell membrane and disperse. This process is known as a “futile cycle.” The “futility” is further enhanced by media flow, which is higher on the periphery of the biofilm. As a result, bacteria prefer to absorb extracellular ammonium, and this preference is stronger at the periphery than in the centre, where the flow is smaller and the resulting loss of ammonium is smaller as well.

In effect, in biofilm metabolism the centre of the colony effectively produces ammonium for the periphery. This process depends on the availability of glutamate: the depletion of glutamate at the centre halts the production of ammonium, which as a result makes it impossible for the peripheral organisms to produce glutamine and stops their growth. This is a strictly causal connection between the metabolic cycles of bacteria in the biofilm, which leads to a co-dependence of individuals within a colony.¹ The change in the production of ammonium at the centre of the biofilm is a direct effect of the depletion of glutamate, and the lack of ammonium causes the halting of growth of cells at the periphery.

2.2 *Signalling for synchronization*

In the second case, a signalling description is the one preferred by researchers (see Prindle et al. (2015); note that this process occurs in parallel to the ammonium one just described). When the bacteria at the centre of the biofilm deplete the available glutamate, the change in the metabolic state of the cell triggers potassium flux through ion channels, resulting in a depolarization of the cell. Prindle et al. (2015) investigate specifically the YugO potassium channel, which is gated by a TrkA protein known to change conformation in response to ATP binding—a process susceptible to the metabolic state of the cell (see Y. Cao et al. 2013). The increase in extracellular potassium further triggers a short-term membrane depolarization in neighboring cells, which actively propagates the potassium—and depolarization—wave throughout the biofilm.

As this wave spreads, it limits the ability of peripheral cells to take up or retain nutrients, altering the underlying metabolic reactions, but not contributing to them. This happens through physical interactions—as glutamate and ammonium are charged molecules, the change in the voltage of the cell membrane impacts the electrical forces mediating interactions between the cell and the nutrients (see e.g., Hulbert and Else 1999). As a result, the growth slows down, and the nutrients can diffuse to the centre of the biofilm. In this case, a communicational perspective is adopted to describe the process. The central and peripheral cells are regarded, respectively, as senders and receivers. The potassium flux is described as a signal. The intermediate cells, which actively propagate depolarization, form a structure that enables signal transfer.

1. Notably, this enhances the adaptive character of the biofilm, an organization which is more likely to occur under environmental stress; see, e.g., Berk et al. (2012).

3 CAUSAL ANALYSIS OF BIOFILM INTERACTIONS

Consider how the processes would be described by three existing causal accounts of signalling—Ross’s distinction of material and signal continuity, Artiga’s framework of non-enabling causes, and through the lens of the organizational account. Figure~1 depicts the structure of the two processes as a causal graph (see Pearl 2022; Woodward 2004).² The figure combines the metabolic process, presented in the top row, and signalling, presented in the bottom row, as they share a starting point (glutamate concentration in the centre of the biofilm) and the destination (the growth of cells in the periphery). Both causal chains can be considered as pathways controlling the growth of the cells at the periphery (drawing on the criteria in Ross (2018), p. 557): they are sequences of causally connected events that (i) offer fine-grained causal control over the variable of periphery growth (as the rate of growth depends on the concentration of the ammonium or the charge of the extracellular environment in a continuous manner, see Liu et al. (2015), Extended Data Figure 1), that (ii) rely on material or signal continuity (I will consider this extensively in subsection 3.1), and (iii) have a fixed order of steps shared across some large domain of systems.³ Importantly, both pathways provide contributing causes (in the sense of Woodward 2004) for the growth of the cells at the periphery (as highlighted by interventions I_2 , I_4 , and I_5 depicted on Figure 1, Liu et al. 2015; Prindle et al. 2015).

3.1 Distinguishing material and signal continuity

Ross explicitly distinguishes between “material” and “signal” continuity in causal pathways (Ross 2021b). The former is her main focus: material continuity requires significant, reliable carry-over of material that is constrained to the causal process (see Ross 2021b, p. 6). The latter remains relatively unspecified, as her definition requires the continuity of an “informational signal” which in turn is described only as an “immaterial” entity (notably in scare quotes, see Ross 2021b, p. 13). Nevertheless, from the perspective of her account, “cell signalling” would describe pathways which lack material continuity, though she also allows for cases where the “material flow” realizes or constrains “signal flow,” and the two processes can be studied through one another (Ross 2021c, pp. 3-4).

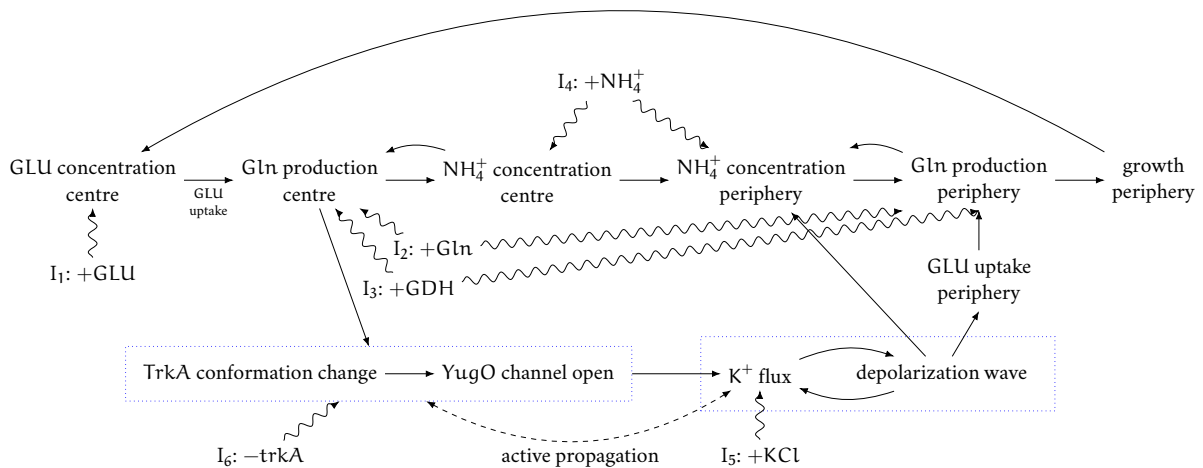
This, in fact, does not help in understanding the distinction visible in the case studies. In both cases we have a form of material continuity: either in propagating changes of ammonium concentration or in the potassium ion flux. Ammonium in the metabolic chain is the direct substrate of reactions at later steps of the pathway, and as such, the physical material gets preserved.⁴ The signalling chain is slightly more interesting.

2. As I am considering type-level biological causes, the resulting graph involves some cycles, a common feature of biological causation (see Bechtel and Bollhagen 2024; Gebharder and Leuridan 2024), which complicate the structure (and more practically, the calculations of probabilities involved, though this is beyond the scope of this paper)—however, these cycles are not vicious in any way, and if we were to consider a token-level description of the same processes, they would unravel (in a time-indexed form).

3. In the case of the metabolic chain, it is likely limited to colonial unicellular organisms, but the signalling chain is conjectured to expand over a broad range of systems as Prindle and colleagues (2015, p. 6) consider “functional similarities with respect to long-range electrical communication,” such as occurs, among others, in nervous systems. These issues connect closely to debates on the representational scope of model organisms (see Ankeny and Leonelli 2020; Prieto and Fábregas-Tejeda 2025).

4. The argument here is analogous to the analysis of glycolysis offered by Ross (2018, pp. 553-555).

Figure 1: Causal graph for biofilm metabolic co-dependence and potassium signalling processes. GLU stands for glutamate, Gln for glutamine, NH_4^+ for ammonium, and K^+ for potassium ions. Nodes connected with wavy arrows depict the interventions introduced in (Prindle et al. 2015; Liu et al. 2015). These included the addition of GLU, Gln, NH_4^+ , GDH (glutamate dehydrogenase, enzyme necessary for the production of free ammonium from glutamate), and KCl to the growth medium, as well as the use of a genetically modified strain of bacteria with the gene coding for the TrkA protein deleted, effectively rendering the YugO channel inoperative. The boxes depicted in the bottom row are introduced to highlight how the description of the process in (Prindle et al. 2015) switches between explanatory levels. For illustration purposes, the diagram is connecting both mechanistic and phenomenal properties of the process.



For Ross, paradigmatic cases of material continuity include processes that can be studied with tracer techniques—through marking entities and tracing the trajectory of the mark, for instance with radioactive materials. This allows researchers to track the subsequent events the marked material is a part of—biochemical reactions, cells it interacts with, etc. These methods are in fact employed by Prindle et al. (2015) to study the role of potassium fluxes. They use voltage-sensitive dyes to record membrane potential and trace the spreading wave of potassium.⁵ This is a more “statistical” kind of continuity compared to examples discussed by Ross, as it does not depend on *individual* molecules but on their concentrations (see also the analysis of molecular-level causation in Godfrey-Smith 2016b). Nonetheless, it is a case of material continuity. The relevant difference-makers are potassium molecules, and the process depends on their constrained carry-over, uncovered with the use of tracers, fulfilling the criteria of material continuity. This conclusion is further supported by the mechanistic dependence of this process on potassium channels in active propagation of the signal (as ascertained by the intervention I_6 depicted on Figure 1, see Prindle et al. 2015). This runs contrary to Ross’s description, which in analogous cases identifies the causes with electrical charge, motivating the delineation of “material” and “signal flow.”

5. Prindle et al. (2015) use Thioflavin T (ThT) and Asante Potassium Green-4 (APG-4). ThT is positively charged and as such can be retained by cells which have negative membrane potential, resulting in serving as a faithful reporter of the membrane potential. APG-4 (an even better exemplar of Ross’s notion of a “tracer”) is used to study the extracellular potassium concentration as it increases fluorescence in response to the potassium cation concentration.

Hence, simply denying material continuity to cell signalling doesn't allow for distinguishing the two cases analyzed here.

3.2 *Non-enabling causes and closure of mechanisms*

Artiga (2021b; 2021a) develops his account specifically for distinguishing signals in organismal communication—both animal, and unicellular. He builds on the standard influence account [Maynard Smith and Harper (2003); see below in subsection 3.3] and the energetic asymmetry between signals and responses (as mentioned in section 1). Artiga develops this view to specifically fit into the neo-mechanistic (Machamer, Darden, and Craver 2000; see also Glennan 2025) mode of explanation, and defines signals as minimal, “non-enabling” causes (Artiga 2021b, p. 8587).

The “ability” of a system to perform a certain action consists in having a mechanism for performing that action. It is a graded notion: “the more parts and activities an entity lacks for performing [an action], the less able it is to [perform that action]” (Artiga 2021b, p. 8590). In this context, an enabling cause is the component or activity that makes the system able to perform the action in question. Signals, on the contrary, as non-enabling causes, require the recipient to be able to execute the response independently. This notion of “minimal cause” is intended to capture what researchers mean when they characterize the role of signals as “triggering,” “inducing,” or “activating.”

Artiga takes the “minimality” strongly, rejecting the role of *any* intrinsic (material and physical) properties of the signal vehicle. This is crucial to account for the causal efficacy of the signal's content on his account. However, this strong interpretation is at odds with the uses of the notion of “signalling” in scientific literature.⁶ Consider the case of potassium signals just described. The molecule's shape, size, and its electric charge are central to its ability to act as a signal (due to how they interact with ion channels and change polarity of cell membranes).

Indeed, this is also true for his analysis of quorum sensing. He notes: “the autoinducer binds receptors CqsS and LuxPQ, which dephosphorylate protein LuxO, via LuxU; since dephosphorylated Lux is inactive, *qrr 1-4* is not transcribed, which causes AphA not to be translated” (Artiga 2021a, p. 6). But dephosphorylation occurs through hydrolysis, and as such depends directly on intrinsic, physical properties of the signal molecules, thus defeating minimalism.

On Artiga's account, these material properties should disqualify the processes as signalling. The involvement of material properties (e.g., molecule size relevant for the channel, its chemical composition relevant for hydrolysis) serves a key explanatory role, contrary to his claim of minimalism. Hence, his account would fail to classify the case of potassium signalling as an instance of signalling, and should fail for quorum signalling as well.

Further, the notion of “non-enabling causes” is unclear when applied at the level of molecular processes. This is due to the statistical nature of causation at the molecular level (see Godfrey-Smith 2016b; Nathan 2014). A sufficient decrease in ammonium concentration triggers the slowdown of metabolism at the periphery similarly to how the increase in potassium concentration does. In both cases the cell retains (some degree of) the ability to produce glutamine, despite one of the molecules being a substrate for metabolism, and the other a “moderator” of

6. A larger catalog of such examples is provided by a digital philosophy of science corpus study reported in Rorot (2025); Rorot (under review).

metabolic processes. Hence, both could be considered as non-enabling causes, even though only the latter is a signalling pathway.

This is relevant for quorum signalling as well, though Artiga highlights that bacterial signals lead to an “exponential increase” in the relevant response. Hence, he concedes that the difference is a matter of degree, but notes that in some cases this is enough for the distinction to hold. However, the concession is more costly than Artiga allows. As the biofilm cases show, it fails precisely when it is needed the most, namely to sort boundary cases, where the differences are small. Hence, the notion of minimal causes doesn’t adequately characterize signals, and the actual role of material properties and the statistical nature of causation at the molecular level leads to paradoxical conclusions for at least some relevant cases.

3.3 Organizational account of biological communication

The organizational account of biological communication laid out by Frick and colleagues (2019) adapts the established “influence” account (e.g., Maynard Smith and Harper 2003; Scott-Phillips 2008) by replacing the etiological concept of function with the organizational one (see Moreno and Mossio 2015). The influence accounts consider signalling as functional influence which a sender exerts on a receiver by means of a signal. This primarily aims to distinguish “signals” from “cues”. Frick et al. continue in this line, defining communication in terms of a self-maintaining organization between the sender and receiver, with the signal triggering a response which maintains the organization, realizing closure of the sender-receiver system.

However, their definition is less specific when it comes to distinguishing signalling and metabolism, a case they explicitly consider:

a signal molecule plays its role *as signal* in virtue of its capability to trigger a regulatory action in the sensory-effector mechanisms of the receiver—and ultimately give rise to a functional influence loop [between sender and receiver, while] a metabolite is used as a substrate (it is consumed) in the metabolic processes of the receiver. Metabolic interactions do not imply a *functional influence* by the sender upon the receiver. They realise *functional couplings* through the exchange of metabolites that are released in the environment and used in the metabolism of the systems involved in the interaction. (Frick, Bich, and Moreno 2019, emphasis original)

The crucial distinction between functional influence and functional coupling is in fact glossed over and cannot adequately delineate the two case studies considered here. The distinction between a metabolite being “used as a substrate” or “consumed” and a signal “triggering a regulatory action” can be read in two ways. Either it points towards the presence of material continuity, and thus the account fails for the same reasons as Ross’s does. Or the distinction between “coupling” and “influence” is the distinction between enabling and non-enabling causes that Artiga relies on, and fails for the same reasons as Artiga’s criterion fails. Either way, their characterization of signalling is not sufficiently detailed to distinguish the boundary case of interest here. The ammonium and potassium interactions unfold concurrently, effectively becoming a single coupled system.

3.4 *Desiderata for criteria of cell signalling*

While the existing causal characterizations of signalling fail to provide sufficient criteria, the foregoing analysis identifies the desiderata that criteria of cell signalling must meet to adequately reflect scientific practice.

First, on the meta-level, the criteria should be descriptively accurate, while preserving a normative bite. They should allow for distinguishing paradigmatic cases of signalling and non-signalling interactions. This includes capturing scientists' intuitions about what signalling is. For instance, contemporary neuroscience depends on the view of neurons as signalling, and the concept of signalling being developed here shouldn't aprioristically reject such cases, but still allow for a debate on the applicability of the concept in such contexts. At the same time, it should allow for identifying cases where the signalling gloss is extended inappropriately. In this sense, the normative aspect of the paper is captured by the notion of "object normativity" (Kaiser 2019), identifying epistemic norms present in scientific practice.

Second, on the object-level, the issue both Ross' and Artiga's characterizations missed is the *material basis of signalling*. This is a broader issue in the literature on biological information, and indeed—in philosophy of information in general. The problem of physical basis of information, including in biology, remains open and relatively unexplored (with the important exceptions of the work by Collier 2011; and Deacon 2007, 2008; Deacon and García-Valdecasas 2023). In the current context, the crucial insight is that information in nature is always carried by a physical vehicle, its physical properties realize the informational structure, and the "immaterial", structural character of information that some theoreticians underscore (e.g., Dodig-Crnkovic 2014; Floridi 2011) appears as a descriptive abstraction.⁷ Hence, defining signalling by denying its material properties is bound to fail, and criteria must allow for some causal role of material properties of signals.

Finally, the criteria for signalling must account for the specific nature of causation at the level of molecular interactions in the cell. Godfrey-Smith (2016b, pp. 485-486) describes it as a "molecular storm" and "unending spontaneous motion", where "the way things get done is by biasing tendencies in the storm, nudging random walks in useful directions, thereby getting a consistent upshot out of vast numbers of mostly meaningless changes." The probabilistic character of causation requires focusing on the larger structure of the process, rather than individual causal interactions, and any interventions need to be considered at scale. This is particularly problematic for graded notions such as Artiga's "enabling" causes, as rarely any individual cellular process will clearly fall at one end of this spectrum, as we saw in the analysis in subsection 3.2.

7. This issue is closely related to the burgeoning debate on medium independence of information processing (e.g., R. Cao 2022; Chirimuuta 2018; Drayson 2025; Garson 2003; Haueis and Colaço 2025; Haugeland 1993; Maley 2023, 2025; Piccinini 2015, 2021; Williams 2025). However, even supporters of medium independence do not deny the realization claim. E.g., Piccinini (2015, p. 123) specifies that information is carried by differences along a single dimension of variability in the medium: "states that the system can distinguish from one another". This limits, but doesn't entirely deny the physicality of information.

4 SIGNALLING AS DETACHMENT

As noted in [section 1](#), the intuitive notion of signalling seems to comprise two elements: the sender-receiver model derived from Shannon, and the idea of removal from immediate physical effects. However, existing accounts fail to provide sufficient criteria to distinguish signalling.

However, both elements can be connected through the concept of “detachment”. Lenny Moss (Moss 2009, p. 107) uses the term to describe “the relative independence of an entity from a larger milieu—its ability to resist the forces of thermal or other kinds of winds.” This idea of causal separation is closely related to the more formal notion of degrees of freedom: a more detached system will have more internal degrees of freedom, its phase space will be of higher dimension. For Moss (2009, p. 106), nature has the tendency to “explore greater levels of detachment”: e.g., molecules are more detached than individual atoms. In the case of biological systems detachment increases rapidly through the inclusion of historical dependencies (e.g., in how future interactions of a protein depend on its conformational history) and group organization, what Moss calls “sociality of detachment.”⁸ While it is a broad and abstract concept, Moss considers examples from genomics, specifically from the expansion of gene families, which pinpoints detachment to specific cellular properties and contributes directly to the distinction of cell signalling.

Moss (2009, p. 111) proposes a thought experiment of two “ideal cell types”, $\mathcal{A}$ and $\mathcal{B}$. Let’s assume $\mathcal{A}$ has a single cell surface receptor, with a wide binding specificity: it responds to red, blue or green cues, but doesn’t distinguish them, producing a unified white output. Cell type $\mathcal{B}$ has six cell surface receptors. Three of those are each specific for red, blue or green features, producing red, blue, and green responses respectively. The other three respond to all three cue types, but each produces a different response type, red, blue, and green, respectively. Hence, when a cell $\mathcal{A}$ senses “blue”, it produces a white response O , whereas cell $\mathcal{B}$ will produce a suite of responses: red, two blues, and green upon receiving the same blue cue. Effectively, cell $\mathcal{A}$ has a lower degree of detachment than cell $\mathcal{B}$. Moss (2009, p. 111) summarizes:

Where cell $\mathcal{A}$ is bound to respond to red, blue or green features of the world in a predictable white way, Cell $\mathcal{B}$ can either act so as to discriminate between the red, blue and green features of the world, or it may act so as to prefigure based upon its own history [...] and] choose to respond in a red way, or in a blue way, or in a green way, to whichever of the features it encounters in its world.

This ability to “choose” responses is precisely what makes Cell $\mathcal{B}$ more detached than $\mathcal{A}$: instead of a single (relevant) degree of freedom $\mathcal{A}$ has, $\mathcal{B}$ has three degrees at its disposal. In this way, detachment also identifies the most interesting, for a researcher, dimensions of the system, specifying the interface between the system and what counts as its environment.

As is clear from the exposition so far, detachment is a graded notion, and thus runs into the same objection raised against Artiga in [subsection 3.2](#). Identifying relevant degrees of freedom

8. Admittedly, his progressive vision of evolution with its claim that “human socio-cultural life-forms are the most detached formations of nature yet” misses the mark. Consider the growing scientific recognition of non-human animal cultures: in primates, cetaceans, birds, or even fish (see the articles in the special issue edited by Brakes et al. 2025; see also Whiten 2021). Further, and in a more direct connection to the current topic, researchers increasingly appreciate the continuity between animal (and possibly even fungi, see Adamatzky 2022) and human communication systems (see recent frameworks in Heintz and Scott-Phillips 2023; Palazzolo 2025; Scott-Phillips and Heintz 2023).

across the two case studies would provide marginal differences between them, and no line between metabolism and signalling. A more formal definition drawn from dynamical systems theory would sharpen the measure but still fail in supplying the demarcation criterion.

However, this objection does not transfer: the degree of detachment is not taken as a criterion here. Detachment is unevenly distributed throughout the process, and signalling is distinguished by its presence at three critical junctures: between the signal vehicle and the response, at the point of reception; between the signal and the machinery by which the receiver generates the response; and between the sender and the receiver, each of which must maintain itself independently for the sender's sensitivity to the response to occur. The degree of detachment is secondary; what matters is whether there is *some* detachment or *none* at the relevant points. Thus, detachment is localizable, and can be directly checked for in any causal process. It underpins the distinction without determining it by amount.

4.1 *Transduction and signal continuity*

Defining signal continuity cannot be done by simply denying material continuity, as I've argued in [subsection 3.1](#). The material structure is relevant for the distinction, and the identification of features along which the system is detached from its environment helps understand what differentiates signal and material continuity.

If we pay attention to the cases of potassium flux in the biofilm and glutamate signalling via transporters, what we note is that while both of these examples involve material continuity, as I have argued above, the potassium case also involves breaks in material continuity. At the same time, as highlighted in the analysis, the standard framework Ross proposes, focused on tracers, is not immediately able to identify this kind of break in continuity. We observe them in two places: at the very beginning of the signalling pathway, where the binding of ATP to the channel protein leads to the depolarization and potassium outflow, and at the very end of the pathway, where the potassium depolarizes the membrane of cells at the periphery preventing them from taking up glutamate from the medium.

Importantly, the discontinuity occurs at the direct cause of the process that we consider the reception of the signal, regardless of whether there are further discontinuities in the process of generating a response or whether it happens exactly once. If we consider a tentative signalling pathway that consists of materially continuous steps to a certain point and materially discontinuous steps from that point onwards, the step marking the division is the point at which the reception of the signal occurs. The downward causal processes are the response to the signal (which is crucial for distinguishing senders and receivers in a number of tightly coupled biological systems). Further, the discontinuities are realized as activities in the signalling mechanism (in the sense of new mechanism, see Machamer, Darden, and Craver 2000; Bechtel and Bollhagen 2021). The material signal (vehicle) is directly involved in the activity, even as the material continuity breaks: the presence of a large number of potassium ions *depolarizes the membrane* which *inhibits the glutamine uptake* in the case of biofilms.

In physical terms this means that cell signalling involves transduction, one form of energy being changed into another. However, the occurrence of transduction is not sufficient to distinguish signalling, as it occurs also when cells pick up informational cues from their environment. While perhaps it is an indicator of informational relationships between biological signals, the distinction of signals and cues is a crucial one (see e.g., Freeberg et al. 2021; Maynard Smith and Harper 2003). Instead, transduction is a characteristic of a communication channel which needs

to be identified independently in the form of a directional causal relation between two entities. This relation requires the two other criteria discussed in detail below: the active sending and reception of the signal, characterized by sender's sensitivity to the response.

4.2 *Active reception and systems' abilities*

Detachment allows for capturing the intuition behind Artiga's notion of "minimal causes". This distinction is intended to highlight that the receiver is capable of producing the response even in the absence of the signal, but Artiga's minimalism is too strong, as highlighted above.

It is true that signals—how they trigger, modulate, or control the responses—have a distinctive role in the mechanism of the response. They are contingent with regards to the response being produced, even if they causally contribute to the actual outcome. This can be captured, again, by considering the responses as detached from the signals. As mentioned above, while detachment is a matter of degree, tracing it allows for identifying the specific dimensions of the system which constitute its interface with what counts as its environment. What characterizes signalling is that this interface includes the mechanisms of signal reception and the production of outcomes, i.e., that responses are detached from signals. This can be considered for minimal switching mechanisms [important in this context for Pattee's framework, see Pattee ([1969] 2012); I come back to this below, in [subsection 4.3](#)]: the signal is what triggers the production of a response, as well as for more complex systems where specific response being triggered depends on what signal was received, as in the thought experiment of "ideal cell types" above.

In the potassium case study, the signal triggers two types of responses, which rely on the same mechanism but have different effects. The first type is the slow down of growth, the second—the active propagation of potassium signal. In both cases the potassium charge influences polarization of the cell membrane. This modulates how it interacts with charged glutamate molecules, and the depolarization itself causes the propagation of the potassium signal, as the cell releases potassium to repolarize. This response is minimally detached, as the cell's ability to maintain a specific charge of the membrane is independent of the potassium signals, however the signal does modulate the charge. Without the signal the membrane would be continuously polarized, whereas upon the reception of potassium it depolarizes rapidly. Artiga's characterization would qualify the potassium as an enabling cause, as the outcome depends on its physical properties—its electrical charge. However the polarization and depolarization mechanism is actively maintained by the recipient cell, specifically, by the activity of its ion channels, and as such it's detached from the signal.

This is clearly visible when we consider the experiments with mutant bacteria with the ion channel fully or partially inoperative (Prindle et al. 2015, marked as I_6 on [Figure 1](#)). These mutations limit the ability of bacteria to send out and amplify potassium signals. However, the mutations have no impact on mechanisms through which cells take up and retain glutamate and ammonium. As a result, the growth is expected to be analogous in the mutant bacteria to those in the wild type.⁹ This indicates that the mechanism which generates the response

9. Growth oscillations in the mutant strain are not reported in either article. This can be considered a prediction of the proposed framework: if the growth oscillations are attenuated in mutant bacteria (as the *trkA* mutation limits, but doesn't fully stop the potassium propagation and the related signal amplitude), the postulate of detachment is incorrect, and some other property motivates the distinction between metabolic and signalling processes. A strain in which growth oscillations tracked signal attenuation too closely, or not at all, would significantly undermine the current account.

in the wild type is detached from signals. On the other hand, if the ammonium limitation is circumvented,¹⁰ the potassium flux subsides—as there is no need for signalling. Hence, signalling is not necessary nor sufficient for the metabolic processes to occur. In effect, the two processes are detached—despite both rendering the notion of enabling causes paradoxical, as discussed before.

4.3 Sender-receiver asymmetries and closure

The final component that is often taken as characteristic of signalling is the asymmetry between the signal and the response. This asymmetry has been classified in different contexts in somewhat contradictory terms: a common view in animal communication, noted in [section 1](#), casts signals as energetically cheap compared to responses they elicit. This is clear in the case of alerting my friend of danger that I opened the paper with: shouting to them that there's a car coming involves significantly lower energy expenditure on my part, than their stepping back on the pavement. However, this characterization is not easily applicable to cell signalling, as visible in case studies: the signal, potassium released by bacteria at the centre, has very similar, if not exactly the same energetic cost, as the response, as both employ similar cellular machinery.

The second kind of asymmetry has been suggested by Howard H. Pattee ([1969] 2012) specifically in relation to cell signalling. Pattee claimed that in signalling, complex interactions lead to a simple result. This motivated him to use switching networks as a model of the origins of biological signals. But for him the problem is tied to the explanation of development and replication, with the genetic code being the paradigmatic example of a highly evolved biological message the framework should eventually explain. In this way, his account zooms out on the broader problem of biological information, which goes beyond the scope of the current problem. Still, Pattee raises a number of directly relevant points: he notes that paradigmatic instances of signals, with genetic code and natural languages as prime examples, exhibit hierarchical organization in which higher levels constrain lower elements in such a way that understanding higher-level dynamics doesn't require a full knowledge of its constitutive elements and their state. The abstraction is precisely what the asymmetry he identifies refers to, emphasizing its epistemic status: “communication is in some way a simplification of a complex dynamical process” (Pattee [1969] 2012, p. 58), from the perspective of its observer—whether that observer is the scientist, or another cell that receives the message.¹¹

10. This is tested through the addition of ammonium or direct supply of glutamine to the medium, marked on [Figure 1](#) as I_4 and I_2 , respectively.

11. The characterization aligns with his concept of the

epistemic cut that appears conceptually inescapable in order to distinguish what we call objective knowledge from the individual subjective records or experience of the observer. [...] the most primitive epistemic cut happened at the origin of life which separated the individual cell's genetic informational constraints from the objective lawful dynamics it controls. (Pattee 2012, p. 5, emphasis original)

The epistemic cut, for Pattee essential to life, plays a significant role in his explanation of the mind-brain problem (which he considers as a species of a more general “symbol-matter” problem, see Pattee [1982] 2012), connecting to the observer problems in quantum physics, and reminiscing the more recent concept of “agential cut” developed by Karen Barad (2007; see Branczio 2023). As such, it deserves a significantly more extensive treatment than the current paper allows for. Pattee himself discusses it extensively in the introduction to his collected papers (Pattee and Rączaszek-Leonardi [1982] 2012).

The abstraction Pattee identifies can also be considered through the concept of detachment for a less anthropomorphic description of cellular activity. It captures an important element of the materially discontinuous step in the signalling pathway in the potassium case: the “interface” created by transduction abstracts away from some material details of the process: specific number of potassium molecules available, their exact position, velocity, etc. The detachment simplifies the process—both for the external researcher, and for the cell itself.

There is also a third kind of asymmetry that the organizational account introduces (Frick, Bich, and Moreno 2019), though implicitly: signalling is necessary to maintain closure since the response of the receiver provides something the sender cannot achieve themselves. The response can be both further signalling, as well as some other form of activity—or both at the same time, as in the case of potassium signalling. But importantly, the sender is sensitive to that response. This sensitivity is what distinguishes the sender, and it is a consequence of the sender’s detachment: only a system that maintains itself independently of the receiver can be said to be served by what the receiver does. In organizational terms, not only is the sender-receiver system organizationally closed, but both its components need to be as well (as already highlighted for the receiver in [subsection 4.2](#)), with asymmetric needs for the higher-level organization. In some cases the different needs are reflected by an energetic asymmetry of the first type discussed here. In all instances the embeddedness of closure of the sender and receiver and their mutual relation leads to the kind of abstraction that Pattee identified as the asymmetry.

As in the case of active reception, here detachment introduces requirements not directly on the causal structure of the signalling process, but instead on the internal structure of the components of that process, and how they relate to the relevant instance of communication.

4.4 *Detachment in B. subtilis biofilm*

To summarize the criteria introduced above, consider again the two case studies.

The potassium wave constitutes a genuine case of signalling, as the signal used is transduced as it arrives at the recipient cells. The response it elicits is produced without the signal vehicle playing a constitutive role, through a mechanism internal to the receiver. Finally, the sender is sensitive to the response: the slow down of growth enables the glutamate to spread through the biofilm, providing a necessary component for the sender to maintain its organizational closure—and effectively, for the biofilm as the whole sender-receiver system to maintain its organizational closure at a higher level. Additionally, the response is also characterized by the complexity asymmetry—from the perspective of the sender, the specific “processing” of the signal is irrelevant, only the switch in metabolism and growth matters.

In the ammonium case the first two criteria are not met, and the relevant process is a metabolic, non-communicative interaction. The signal is fully materially continuous, with no transduction occurring, and the response is constitutively dependent on the signal (this is slightly awkward, as the relevant signal is an absence, yet the dependency does hold). The third criterion is met, quite obviously, as the relevant entities are the same ones as in the potassium case. Hence, the asymmetry criterion in any form, on its own, is not sufficient for the distinction.

However, the third criterion comes into play in a number of other cases. Consider how plants, when preyed upon by herbivores, in some instances release chemicals, Volatile Organic Compounds (VOCs), that other plants in the vicinity pick up. This allows the recipients to produce deterrents, so when the herbivores eventually move onto them, they have greater chances of repelling the predator and surviving. This case is considered genuine communication

by a number of scientists (e.g., Baluška and Ninkovic 2010). However, this case fails the third criterion set out here: there is no clear sensitivity to response on the part of the sender. As such, VOCs more likely present a case of natural signs, or cues that the recipients pick out from the environment (though this doesn't deny that other interactions within and between plants still can constitute signalling). Similarly, the case of airborne disease, mentioned in section 1, fails at least the third criterion as well (though it likely also fails the transduction criterion, considering material continuity in e.g., viral infections). In such cases, we cannot consider an infected individual as a sender, as the relevant asymmetry characterized by the sensitivity to response does not hold.

The three criteria are jointly sufficient as well as individually necessary—the causal separation at three points identified is what distinguishes biological signalling. Signalling requires the signal to be transduced, the response to be produced fully by the receiver, and the sender to be sensitive to the outcome of the process. If any of those is not met, the process in question is not communication. Metabolism fails the transduction criterion, as the transmitted material becomes a direct physical substrate of the process, and the active reception criterion, as the recipients cannot sustain their metabolism without received molecules. Cues fail the asymmetry criterion: the senders do not care about the outcomes.

5 IMPLICATIONS BEYOND CELL SIGNALLING

The detachment view of biological signalling impacts several debates in philosophy of biology, most importantly on biological information (e.g., Callebaut and Collier 2006; Godfrey-Smith 2007; Maynard Smith 2000; Stegmann 2016). Covering this topic in full would go well beyond the scope of this article, hence I will focus on two problems influenced most directly: the role of sender-receiver models, and the relation between information, causal specificity, and the concept of control in biological systems.

5.1 *Sender-receiver models*

Emerging from the work of David Lewis (Lewis 1969), sender-receiver models have been established as part of the signalling games framework in the work of Brian Skyrms (Skyrms 2010), who connected them explicitly to information theory (see the review in Zubek, Korbak, and Rączaszek-Leonardi 2024). They have been used to study animal communication (e.g., Godfrey-Smith 2012, 2013; Mann 2018), quorum sensing in bacteria (Pacheco et al. 2015), as well as information in genes and in memory (Godfrey-Smith 2014; Bergstrom and Rosvall 2011; Shea, Godfrey-Smith, and R. Cao 2018). Briefly, the simplest versions of these models (Lewis 1969) assume a sender and receiver with asymmetric access to the world (sender sees the stimulus the receiver must react to) and with shared interest (an equal reward is paid to both, conditioned on the receiver's "action"). The close relation between signalling and broader evolutionary games allows for studying informational properties of interactions while preserving the evolutionarily relevant characteristics of those interactions.

The account developed here identifies criteria for identifying the extent of idealizations in sender-receiver models, specifying the limits of their interpretability. In this context, sender-receiver models can be understood as a hypothesis regarding the existence of a signalling interaction within the target process. Tracing detachment in the target is necessary for a full

understanding of how such models relate to the actual biological phenomena. This responds to some of the charges against the informational views presented by supporters of the influence definitions of signalling (e.g., Owren, Rendall, and Ryan 2010; Rendall, Owren, and Ryan 2009) and provides a clear guide on distinguishing instrumentalist and realist uses of information theory in biological contexts (in response to e.g., Levy 2011; Sarkar 1996).

5.2 *Causal specificity and control*

Causal specificity (Woodward 2010, p. 305) identifies processes of cause and effect where “C will influence E to the extent that by varying the state of C and its time and place of occurrence, we can modulate the state of E in a fine-grained way” (under a given description). In reference to biology, the concept has been invoked to account for the characteristics of DNA: a change in DNA sequence determines what RNA and protein sequences will be produced in a highly narrow fashion, as contrasted to e.g., RNA polymerase (Waters 2007), which play a more generic role in the production of proteins. While this example is not uncontested (Oyama 2000; P. E. Griffiths and Gray 1994; Stotz and P. E. Griffiths 2017), the idea of some causes being more specific than others is relatively natural to biologists.

Causal specificity is closely tied to the concept of control (Stegmann 2014; Winning and Bechtel 2018), as causes with greater specificity are considered as exercising causal control over the processes they influence. Further, specificity is tied to information (e.g., Weber 2006), as more specific causes carry more information for the process they influence.

Moss’s thought experiment of ideal cell types (see section 4) clearly connects higher levels of detachment to greater causal specificity. However, the concept does not collapse into just specificity, and not all cases of causal specificity involve detachment. Consider self-organizing systems such as Rayleigh-Bénard cells, a regular pattern of convection cells occurring in a fluid heated from below. This requires a relatively specific relation between the heat source and the fluid, as varying the temperature, direction, location of the source can modify the pattern of the cells in a fine-grained manner. However, there is no detachment between the pattern and the heat source: if the heat is turned off, the pattern dissipates.

The concept of detachment is thus better equipped than specificity itself to explain biological control. Winning and Bechtel (2018) note the insufficiency of new mechanism for accounting for biological control. Most importantly, they point out that mechanists fail to explain where capacities of mechanisms come from. Thus, they fail to account for mechanisms’ activities which are traditionally defined as exercise of those capacities. To solve this problem, they propose to ground causal capacities in the concept of constraints (originally introduced in this context by Juarrero 1999; it is also central for Deacon 2011), and note:

The tendency or capacity to resist, re-route, displace, etc. various forces is just what it is to be a causal power. Thus, on our view, when constraints enable objects to have novel, emergent behaviors, this is tantamount to the emergence of causal powers. (Winning and Bechtel 2018, p. 294)

Their account of causal powers aligns closely with the concept of detachment. However detachment, as understood here, is an extrinsic property of mechanisms, while constraints as causal powers are constitutive of mechanisms, and so, intrinsic. This allows detachment to more directly explain biological autonomy, as indicated in the discussion of closure, which remains

an open problem for Winning and Bechtel (2018). Most importantly, just as detachment allows for identifying the relevant sender and receiver within a process, it also allows for identifying controller-plant (to use the cybernetic term) relations, as indexed to a particular instance of control. This is somewhat unsurprising given that signalling is almost universally involved in some form of control. Its relationship to causal specificity and information accounts for the immense capabilities of complex signalling systems for control, as exhibited by central nervous systems of animals.

Importantly, Calcott and colleagues (2020) connect the three strands explicitly. Building on Skyrms' sender-receiver models viewed as causal networks, they consider how specificity emerges in signalling and highlight its connection to control in an interventionist framework. However, their account aims to provide an informational description of the process, and the primary achievement they claim is that the account covers the evolution of informational signals. However, as already briefly noted, their account presumes the occurrence of a signalling structure (it does so by adopting the Skyrmsian framework, see [subsection 5.1](#)), and as such is oriented towards a different problem than the account proposed here.

6 CONCLUSIONS

The goal of the paper was to identify criteria that distinguish biological signalling from other types of interactions at the level of cells. Building on two sets of experiments on *B. subtilis* biofilms, I have shown where existing accounts fail to appropriately account for scientific practice, and suggested that the concept of detachment introduces a theoretically grounded way of improving and connecting the existing criteria. Detachment is a property of processes and mechanisms that exhibit a degree of independence from their environment, tracing causal separation within and between processes. It characterizes signalling in three ways. First, the signal itself is transmitted materially between the sender and receiver in such a way that it is transduced at the point of reception: signals are detached from the responses they elicit. Second, the receiver is detached from the sender and from their environment in such a way that it is capable of actively producing the response independently of signalling. Third, the sender is sensitive to the recipient's response.

Reconceptualizing the concept of signalling impacts the ongoing debates on biological information. For sender-receiver models, the proposed criteria allow for specifying the scope of idealizations such models involve in any given instance. Detachment also captures the requirements for biological autonomy, impacting debates on biological control—including its complex forms that are the focus of neuroscience. Importantly, the criteria supplied by the concept of detachment in no way preclude the applicability of existing work on information flow in signalling systems (Barwise and Seligman 1997; Calcott, Pocheville, and P. Griffiths 2020). These criteria provide what these other accounts assume: a way of identifying which causal structures are adequately characterized as signalling, and thus are amenable to informational description.

Further, the concept of detachment highlights the complex relation between signalling—whether characterized in informational terms or not—and its material basis. Hence, the criteria laid out here, particularly the role and limits of material continuity in signalling, connect to the ongoing debates on the physical basis of information (e.g., Collier 2008; Deacon 2007, 2008). This topic is particularly relevant for the debates on physical computation in the brain (Chirimuuta 2024; Godfrey-Smith 2016b; Harrison, Rorot, and Laukaityte 2022; Haueis and

Colaço 2025; Miłkowski 2013; Piccinini 2020; Rorot 2022; Seth 2025). As the signalling language is closely related to computational explanations of brain function, the proposed account underscores the relevance of the material structure of computations carried out by the nervous system. It also promises to impact debates in other areas of philosophy of biology where material constraints are relevant, but escape standard frameworks—one example being the debate whether material overlap is definitional for reproduction (e.g., Godfrey-Smith 2016a; J. R. Griesemer 2005; J. Griesemer 2016).

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